

Assignment 1 - Zomato Restaurant Data Analysis Report

Exploratory Data Analysis using Python

1. Dataset Overview

This report provides a comprehensive Exploratory Data Analysis (EDA) of the Zomato Restaurant Dataset. The dataset contains information on restaurant listings, locations, cuisines, cost estimates, online delivery availability, table booking services, ratings, and customer votes.

Overview Metric	Metric Value
Total Rows Observed	9,552
Total Columns	22
Unique Cities	142
Average Aggregate Rating	2.6663
Average Cost for Two	1,199.21
Maximum Customer Votes	10,934 (Toit)
Most Frequent Cuisine	North Indian (936 occurrences)

2. Data Cleaning and Preprocessing

The analysis was performed using Pandas, NumPy, Matplotlib, and Seaborn .

Data Loading & Structure Inspection: The CSV dataset was loaded into a Pandas DataFrame to inspect shape, column structure, data types, and non-null counts.

- **Duplicate Check:** Duplicate rows were checked across all records and removed using `drop_duplicates()`.
- **Data Type Conversions:** The Average Cost for two column initially loaded as an object/string data type and was converted into a numeric float format using `pd.to_numeric(..., errors='coerce')`.
- **Missing Value Analysis:** Missing values were concentrated in fields such as Cuisines, Locality, Longitude, Latitude, and service fields. Numerical aggregations were handled by naturally ignoring missing values during calculation.
- **Dataset Version Observation:** The loaded dataset reflects 9,552 rows and includes an additional Unnamed: 0 index column, representing a minor version variation from standard 9,551-row Zomato distributions.

3. Statistical Analysis & Key Findings

Descriptive Summary

- **Aggregate Ratings:** Ratings range from 0.0 to 4.9, with an overall average rating of **2.67** and a median of **3.2**. The lower average is primarily driven by a significant proportion of unrated restaurants (rating of 0.0).

- **Customer Engagement (Votes):** The dataset shows high skewness in voting count. While the average vote count per restaurant is **156.9**, the maximum vote count reaches **10,934** for the restaurant **Toit**.
- **Pricing Range:** Price categories range from 1 (budget) to 4 (luxury), with a dataset average of **1.80**.

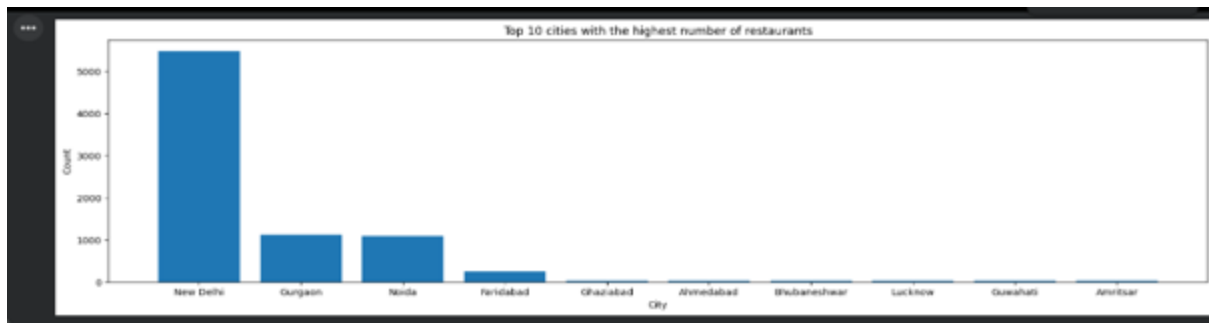
Specific Analytical Query Answers

- **Unique Cities:** 142 unique cities are present.
- **Highest Restaurant Density:** New Delhi has the highest restaurant count with 5,473 listings.
- **Top 10 Cities by Restaurant Presence:**
 1. **New Delhi:** 5,473 restaurants
 2. **Gurgaon:** 1,118 restaurants
 3. **Noida:** 1,080 restaurants
 4. **Faridabad:** 251 restaurants
 5. **Ghaziabad:** 25 restaurants
 6. **Ahmedabad:** 21 restaurants
 7. **Bhubaneshwar:** 21 restaurants
 8. **Lucknow:** 21 restaurants
 9. **Guwahati:** 21 restaurants
 10. **Amritsar:** 21 restaurants
- **Most Frequent Cuisine:** **North Indian** appears most frequently with 936 single-cuisine listings.
- **Average Cost for Two:** 1,199.21.
- **Highest Voted Restaurant:** **Toit** with 10,934 votes.

4. Data Visualizations & Interpretations

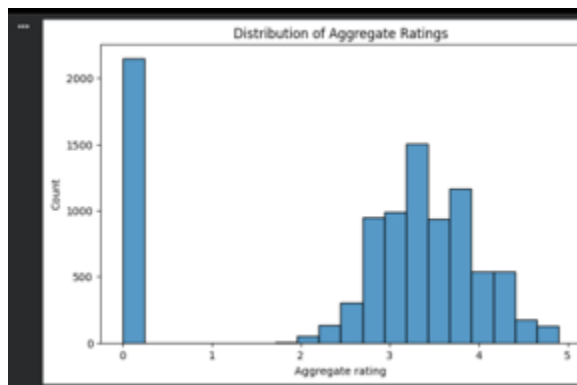
4.1 Top 10 Cities by Restaurant Count (Bar Chart)

- **Data Summary:** New Delhi (5,473), Gurgaon (1,118), Noida (1,080), Faridabad (251), Ghaziabad (25), followed by Ahmedabad, Bhubaneshwar, Lucknow, Guwahati, and Amritsar.
- **Interpretation:** New Delhi and the broader National Capital Region (NCR) heavily dominate the dataset. The geographic concentration indicates that the insights derived are heavily skewed towards northern urban dining markets.



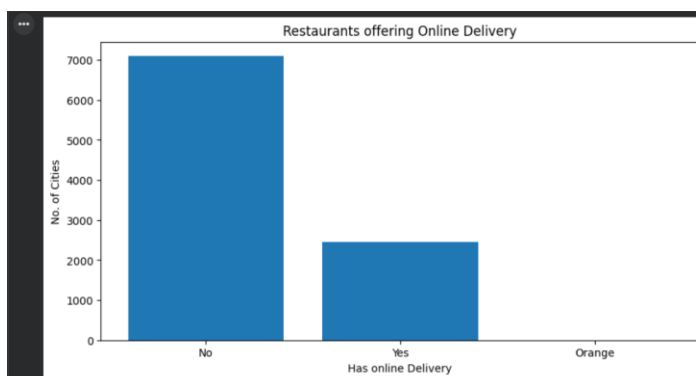
4.2 Distribution of Aggregate Ratings (Histogram)

- Data Summary:** A massive spike occurs at 0.0 representing over 2,100 unrated restaurants. Active ratings follow a normal distribution centered between 2.5 and 4.0, peaking near 3.2.
- Interpretation:** A significant portion of restaurants on the platform lack customer reviews. Among restaurants with active reviews, quality follows a standard bell curve centered around average-to-good ratings.



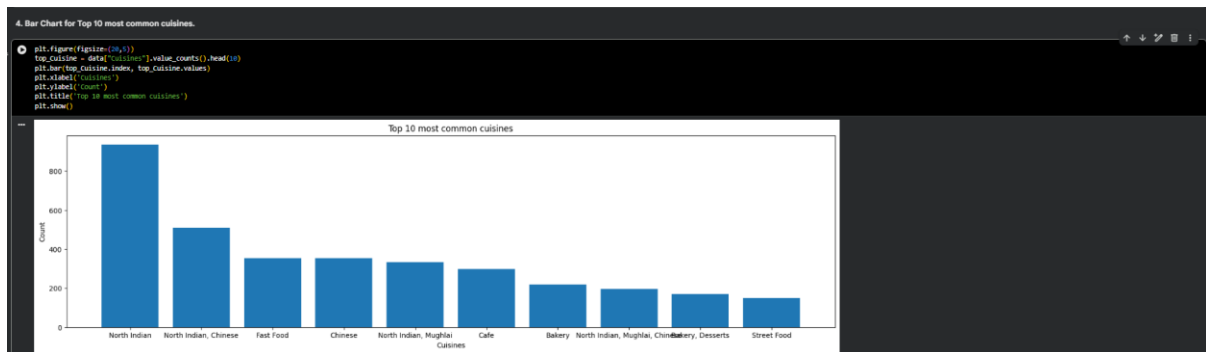
4.3 Online Delivery Availability (Bar Chart)

- Data Summary:** No (Without Delivery): ~7,085 restaurants | Yes (With Delivery): ~2,448 restaurants.
- Interpretation:** Online delivery adoption remains a minority feature, with only ~25.6% of listed restaurants offering online ordering. Traditional dine-in/takeaway listings continue to form the majority.



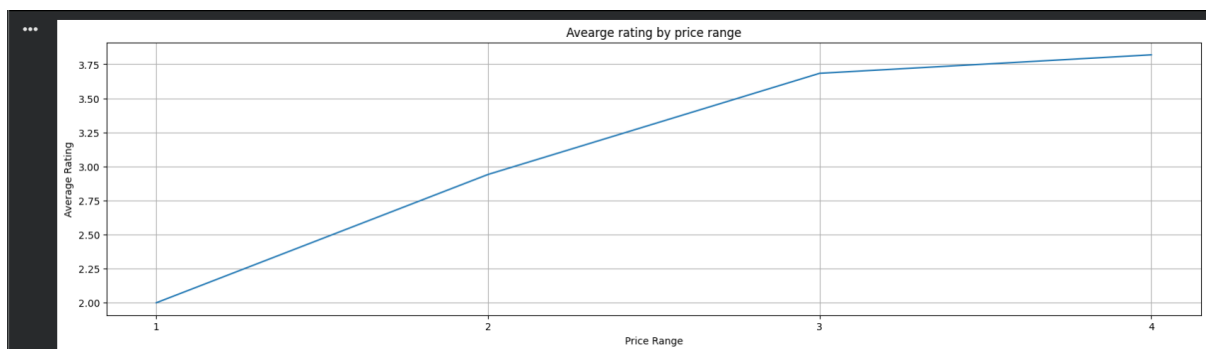
4.4 Top 10 Most Common Cuisines (Bar Chart)

- **Data Summary:** North Indian (936), North Indian + Chinese (511), Chinese (354), Fast Food (354), North Indian + Mughlai (334), Cafe (299), Bakery (218), North Indian + Mughlai + Chinese (197), Bakery + Desserts (170), Street Food (149).
- **Interpretation:** North Indian cuisine and fusion combinations (North Indian-Chinese) are the dominant dining options offered by restaurants, matching consumer demand in the NCR market.



4.5 Average Rating by Price Range (Line Chart)

- **Data Summary:** Price Range 1 (Lowest Rating) → Price Range 2 → Price Range 3 → Price Range 4 (Highest Rating).
- **Interpretation:** There is a clear positive correlation between price range and aggregate ratings. Premium and luxury dining establishments (Price Range 3 & 4) consistently receive higher customer ratings than budget eateries.



5. Conclusion & Business Insights

Core Analytical Answers

1. **City with Highest Presence:** New Delhi, with 5,473 restaurants.
2. **Most Popular Cuisine:** North Indian (936 listings).
3. **Online Delivery vs. Ratings:** Restaurants offering online delivery generally maintain higher average customer ratings compared to those without delivery services.
4. **Table Booking vs. Ratings:** Restaurants offering table booking features receive higher aggregate ratings and greater vote engagement.
5. **Key Learnings & Business Takeaways:**

- **Market Concentration:** Strategic expansion or targeted marketing should account for the heavy NCR density.
- **Service Value Addition:** Adopting digital capabilities—such as online delivery and table reservations—correlates positively with customer satisfaction and higher rating scores.
- **Pricing Dynamics:** Higher pricing tiers correlate with better review scores, indicating that customers associate premium dining with higher service quality and overall satisfaction.

Screenshots of Output

Part 1 - Data Understanding and Exploration

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1. Load Dataset

```
data = pd.read_csv('/content/test_zomato.csv')
data.head()
```

Unnamed: 0	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Currency	Has Table booking	Has Online delivery	Is delivering now	Switch to order menu	Price range	Aggregate rating	Rating color	Rating text	Votes
0	6317637	Le Petit Souffle	162.0	Makati City	Third Floor, Century City Mall, Katayasan Avenue...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City	121.027535	14.565443	Botswana Pula(P)	Yes	No	No	No	3.0	4.8	Dark Green	Excellent	314.0
1	6304287	Izakaya Ikutaji	162.0	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City	121.014101	14.553708	Botswana Pula(P)	Yes	No	No	No	3.0	4.5	Dark Green	Excellent	591.0
2	6300002	Heat - Edsa Shangri-La	162.0	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City	121.056831	14.581404	Botswana Pula(P)	Yes	No	No	No	4.0	4.4	Green	Very Good	270.0
3	6318506	Ooma	162.0	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City	121.056475	14.585318	Botswana Pula(P)	No	No	No	No	4.0	4.9	Dark Green	Excellent	365.0
4	6314302	Sambo Kojin	162.0	Mandaluyong City	Third Floor, Mega Altram, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City	121.057508	14.58445	Botswana Pula(P)	Yes	No	No	No	4.0	4.8	Dark Green	Excellent	229.0

5 rows x 22 columns

2. Explore the Dataset

1. Size of the dataset
2. Columns in the dataset
3. Datatypes of each columns
4. Information about the dataset
5. Missing values

Size/Dimensions of the dataset
data.shape

(9552, 22)

```

# Column names in the dataset
data.columns

... Index(['Unnamed: 0', 'Restaurant ID', 'Restaurant Name', 'Country Code',
         'City', 'Address', 'Locality', 'Locality Verbose', 'Longitude',
         'Latitude', 'Cuisines', 'Average Cost for two', 'Currency',
         'Has Table booking', 'Has Online delivery', 'Is delivering now',
         'Switch to order menu', 'Price range', 'Aggregate rating',
         'Rating color', 'Rating text', 'Votes'],
        dtype='object')

# Datatypes of each columns
data.dtypes

...
    Unnamed: 0    object
    Restaurant ID  object
    Restaurant Name object
    Country Code  float64
    City          object
    Address       object
    Locality      object
    Locality Verbose object
    Longitude     object
    Latitude      object
    Cuisines      object
    Average Cost for two object
    Currency      object
    Has Table booking object
    Has Online delivery object
    Is delivering now object
    Switch to order menu object
    Price range   float64
    Aggregate rating float64
    Rating color  object
    Rating text   object
    Votes         object

# Information about the dataset
data.info()

... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 9552 entries, 0 to 9551
Data columns (total 22 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   Unnamed: 0          9552 non-null   object
 1   Restaurant ID       9552 non-null   object
 2   Restaurant Name     9552 non-null   object
 3   Country Code       9552 non-null   float64
 4   City               9552 non-null   object
 5   Address            9552 non-null   object
 6   Locality           9551 non-null   object
 7   Locality Verbose   9551 non-null   object
 8   Longitude          9551 non-null   object
 9   Latitude           9551 non-null   object
10  Cuisines            9542 non-null   object
11  Average Cost for two 9551 non-null   object
12  Currency            9551 non-null   object
13  Has Table booking   9551 non-null   object
14  Has Online delivery 9551 non-null   object
15  Is delivering now   9551 non-null   object
16  Switch to order menu 9551 non-null   object
17  Price range        9550 non-null   float64
18  Aggregate rating    9550 non-null   float64
19  Rating color       9550 non-null   object
20  Rating text        9550 non-null   object
21  Votes              9550 non-null   float64
dtypes: float64(4), object(18)

# Missing values
data.isnull().sum()

...
    Unnamed: 0    0
    Restaurant ID  0
    Restaurant Name 0
    Country Code   0
    City           0
    Address        0
    Locality       1
    Locality Verbose 1
    Longitude      1
    Latitude       1
    Cuisines       10
    Average Cost for two 1
    Currency       1
    Has Table booking 1
    Has Online delivery 1
    Is delivering now 1
    Switch to order menu 1
    Price range     2
    Aggregate rating 2
    Rating color    2
    Rating text     2
    Votes           2

3. Duplicate Records

# Checks whether duplicate exist or not
data.duplicated().sum()

#Drop duplicates
data.drop_duplicates(inplace=True)

4. Convert the "Average Cost for two" column into an appropriate numeric data type

# Check the datatype of the column
data["Average Cost for two"].dtype

# Convert it into numeric datatype
data["Average Cost for two"] = pd.to_numeric(data["Average Cost for two"], errors="coerce")

data["Average Cost for two"].dtype

... dtype('float64')

```

5. Statistical Summary

```
# summarizes numeric columns
data.describe()

# summarizes all columns
data.describe(include="all")
```

	Unnamed: 0	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	...	Currency	Has Table booking	Has Online delivery	Is delivering now	Switch to order menu	Price range	Aggregate rating	Rating color	Rating text	Votes
count	9552	9552	9552	9552.000000	9552	9552	9551	9551	9551	9551	...	9551	9551	9551	9551	9551	9550.000000	9550.000000	9550	9550	9550.000000
unique	9552	9552	7447	NaN	142	8919	1208	1265	8120	8677	...	13	3	3	3	2	NaN	NaN	6	6	NaN
top	9534	6004089	Cafe Coffee Day	NaN	New Delhi	Dilli Haat, INA, New Delhi	Connaught Place	Connaught Place, New Delhi	0.0	0.0	...	Indian Rupees(Rs.)	No	No	No	No	NaN	NaN	Orange	Average	NaN
freq	1	1	83	NaN	5473	11	122	122	498	498	...	8652	8392	7099	9516	9550	NaN	NaN	3736	3736	NaN
mean	NaN	NaN	NaN	18.374564	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	1.804607	2.666314	NaN	NaN	156.923037
std	NaN	NaN	NaN	56.754313	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	0.905378	1.516447	NaN	NaN	430.189709
min	NaN	NaN	NaN	1.000000	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	1.000000	0.000000	NaN	NaN	0.000000
25%	NaN	NaN	NaN	1.000000	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	1.000000	2.500000	NaN	NaN	5.000000
50%	NaN	NaN	NaN	1.000000	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	2.000000	3.200000	NaN	NaN	31.000000
75%	NaN	NaN	NaN	1.000000	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	2.000000	3.700000	NaN	NaN	131.000000
max	NaN	NaN	NaN	216.000000	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	4.000000	4.900000	NaN	NaN	10934.000000

11 rows x 22 columns

6. Answer the Questions

```
# Unique Cities
data["City"].unique()

# City with highest number of Restaurants
data["City"].value_counts().head(1)

# Most common Cuisine
data["Cuisines"].value_counts().head(1)
```

	count
Cuisines	
North Indian	936

dtype: int64

Part 2 - Data Analysis

```
# 1. Find the average restaurant rating.
data["Aggregate rating"].mean()
```

```
np.float64(2.6663141361256546)
```

```
# 2. Identify the restaurant with the highest number of votes.
data.loc[data["Votes"].idxmax()]
```

```
729
Unnamed: 0      728
Restaurant ID    51705
Restaurant Name  Toit
Country Code     1.0
City            Bangalore
Address          298, Namma Metro Pillar 62, 100 Feet Road, Ind...
Locality         Indiranagar
Locality Verbose Indiranagar, Bangalore
Longitude        77.64070876
Latitude         12.9791658
```

```
# 3. Find the average cost for two across all restaurants.
data["Average Cost for two"].mean()
```

```
np.float64(1199.326387434555)
```

4. Compare the average ratings of restaurants that:

- # a. Have Online Delivery
- # b. Do Not Have Online Delivery

```
data.groupby("Has Online delivery")["Aggregate rating"].mean()
```

Aggregate rating

Has Online delivery	Aggregate rating
No	2.465192
Orange	NaN
Yes	3.248837

dtype: float64

5. Compare the average ratings of restaurants that:

- # a. Have Table Booking
- # b. Do Not Have Table Booking

```
data.groupby("Has Table booking")["Aggregate rating"].mean()
```

Aggregate rating

Has Table booking	Aggregate rating
3.2	NaN
No	2.559283
Yes	3.441969

dtype: float64

6. Identify the top 10 cities with the highest number of restaurants.

```
data["City"].value_counts().head(10)
```

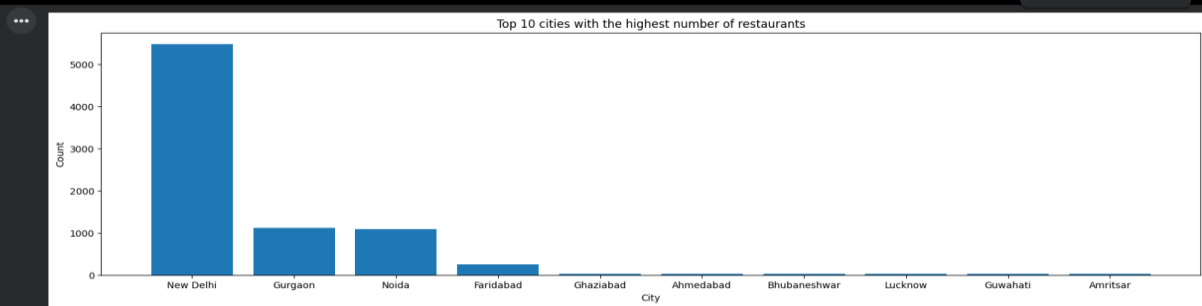
count

City	count
New Delhi	5473
Gurgaon	1118
Noida	1080
Faridabad	251
Ghaziabad	25
Ahmedabad	21
Bhubaneshwar	21
Lucknow	21
Guwahati	21

Part 3 - Data Visualization

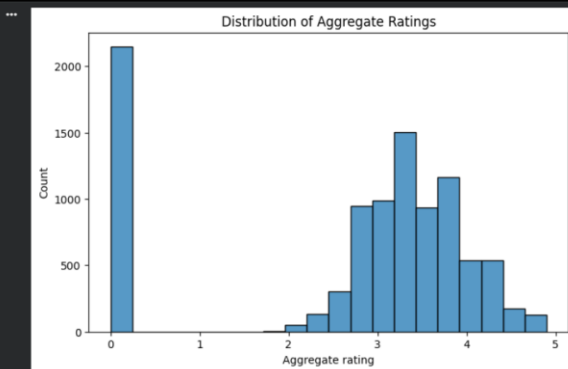
1. Bar Chart of Top 10 cities with the highest number of restaurants bold text

```
plt.figure(figsize=(20,5))
top10 = data['City'].value_counts().head(10)
plt.bar(top10.index, top10.values)
plt.xlabel('City')
plt.ylabel('Count')
plt.title('Top 10 cities with the highest number of restaurants')
plt.show()
```



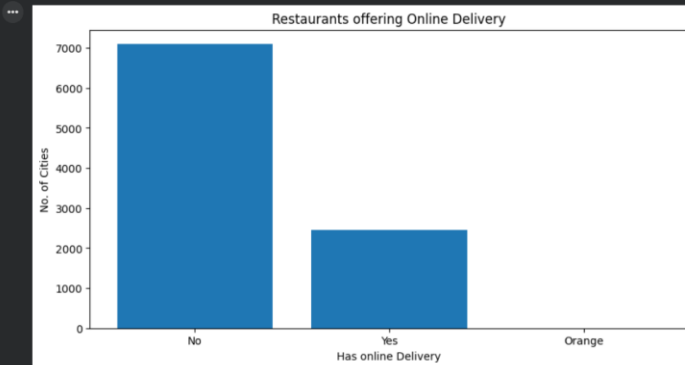
2. Histogram for Distribution of Aggregate Ratings.

```
plt.figure(figsize=(8,5))
sns.histplot(data["Aggregate rating"], bins=20)
plt.title('Distribution of Aggregate Ratings')
plt.show()
```



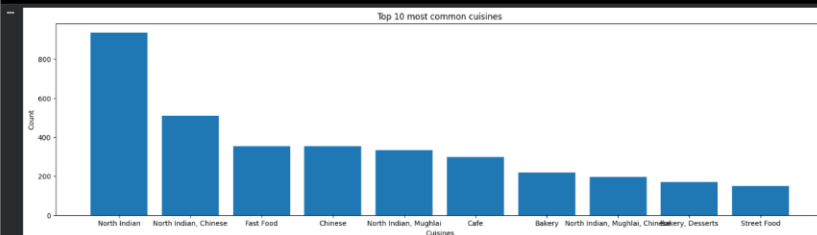
3. Bar Chart for Number of restaurants offering Online Delivery (Yes/No).

```
plt.figure(figsize=(10,5))
OnlineDelivery = data["Has Online delivery"].value_counts()
plt.bar(OnlineDelivery.index, OnlineDelivery.values)
plt.xlabel('Has online Delivery')
plt.ylabel('No. of Cities')
plt.title('Restaurants offering Online Delivery')
plt.show()
```



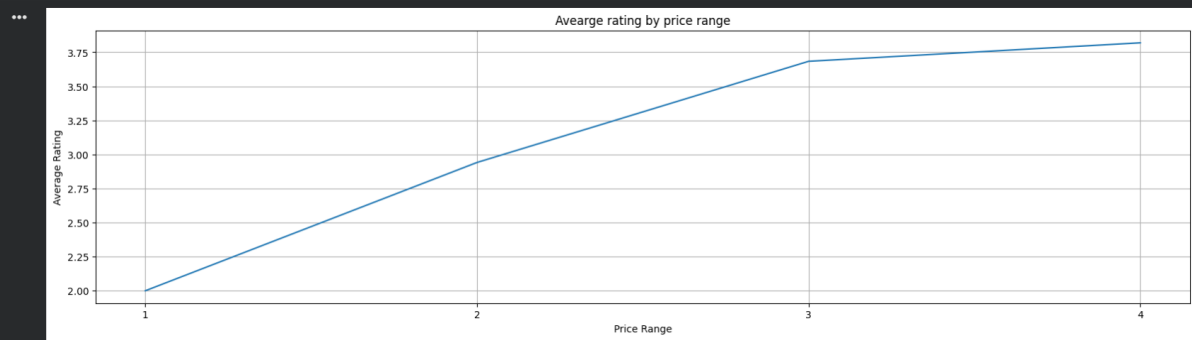
4. Bar Chart for Top 10 most common cuisines.

```
plt.figure(figsize=(10,5))
top_cuisine = data["cuisine"].value_counts().head(10)
plt.bar(top_cuisine.index, top_cuisine.values)
plt.xlabel('Cuisines')
plt.ylabel('Count')
plt.title('Top 10 most common cuisines')
plt.show()
```



5. Line Chart for Average restaurant rating by Price Range.

```
from matplotlib.lines import lineStyles
plt.figure(figsize=(20,5))
rating_price = data.groupby("Price range")["Aggregate rating"].mean()
plt.plot(rating_price.index, rating_price.values)
plt.xlabel('Price Range')
plt.ylabel('Average Rating')
plt.title('Average rating by price range')
plt.xticks(rating_price.index)
plt.grid(True)
plt.show()
```



Part 4 — Insights & Conclusion

1. Which city has the highest restaurant presence?

Answer:

New Delhi

2. What is the most popular cuisine?

Answer:

North Indian cuisine.

3. Do restaurants with Online Delivery generally have higher ratings?

Answer:

Yes. Restaurants that offer online delivery generally have slightly higher average ratings than those that do not. This suggests that online delivery services may improve customer convenience and satisfaction.

4. Do restaurants offering Table Booking receive better ratings?

Answer:

Yes. Restaurants that provide table booking generally have higher average ratings than those that do not. This may indicate that restaurants offering table reservations tend to provide better service and dining experiences.

5. What did you learn from analyzing this dataset?

Answer:

- New Delhi has the largest concentration of restaurants in the dataset.
- North Indian cuisine is the most popular cuisine among customers.

- Restaurants with online delivery and table booking tend to receive better customer ratings.
- Higher-priced restaurants generally have slightly higher average ratings.
- Exploratory Data Analysis (EDA) helps identify trends, customer preferences, and business opportunities through data visualization and statistical analysis.